

§1. Introduction

§2. Main result

§3. Sketch of Proof

§1Question 1Classify all $f(x)$ of the form

$$f(x) := \sum_i C_i \left\lfloor \frac{x}{m_i} \right\rfloor \quad (C_i \in \mathbb{Z}, m_i \in \mathbb{N})$$

such that $f(x) \in \{0, 1\}$ for $\forall x \in \mathbb{R}$ Question 2Classify $a = (a_1, \dots, a_r)$, $b := (b_1, \dots, b_s)$ with $a_i, b_i \in \mathbb{N}$ such that

$$u_n(a, b) := \frac{(a_1 n)! \cdots (a_r n)!}{(b_1 n)! \cdots (b_s n)!} \in \mathbb{Z}$$

for $\forall n \in \mathbb{N}$.Theorem 1 [Bober '09]

For $(m_1, \dots, m_{2N+1}) \in \mathbb{N}^{2N+1}$ with $\gcd(m_1, \dots, m_{2N+1}) = 1$, $m_i \neq m_j$ ($1 \leq i \leq N$, $N+1 \leq j \leq 2N+1$),

$$f(x) := \sum_{i=1}^N \left\lfloor \frac{x}{m_i} \right\rfloor - \sum_{j=N+1}^{2N+1} \left\lfloor \frac{x}{m_j} \right\rfloor.$$

Then $f(x) \in \{0, 1\}$ for $\forall x \in \mathbb{R}$ $\iff$ $f(x)$ is one of

(i) 5 infinite families (e.g.) $f(x) = \left\lfloor \frac{x}{ab} \right\rfloor - \left\lfloor \frac{x}{a(a+b)} \right\rfloor - \left\lfloor \frac{x}{b(a+b)} \right\rfloor$

$$(a, b \in \mathbb{N}, \gcd(a, b) = 1)$$

(ii) 52 sporadic functions given by

$$(m_1, \dots, m_N) := \left(\frac{L}{a_1}, \dots, \frac{L}{a_N} \right)$$

$$(n_{2N+1}, \dots, n_{2N+1}) := \left(\frac{L}{b_1}, \dots, \frac{L}{b_{N+1}} \right),$$

where $L := \text{lcm}(a_1, \dots, a_N, b_1, \dots, b_{N+1})$.

Theorem 2 [Bober]

$a = (a_1, \dots, a_N)$, $b = (b_1, \dots, b_{N+1})$, $a_i, b_j \in \mathbb{N}$.

Then $u_n(a, b) \in \mathbb{Z}$ for $\forall n \in \mathbb{N}$.

$\Leftrightarrow (a, b)$ is one of

(i) 3 infinite families (e.g.) $a = (a+b)$, $b = (a, b)$ ($a, b \in \mathbb{N}$, $\gcd(a, b) = 1$)

(ii) 52 sporadic (a, b) .

Idea of Proof

Reduce the classification to that of "algebraic hypergeometric functions".

Remark 1

Theorem 1 is conjectured by Vosyuniin [1999].

Definition

$d_1, \dots, d_r, \beta_1, \dots, \beta_s \in \mathbb{Q}$, $\beta_j \notin \mathbb{Z}_{\leq 0}$.

$${}_rF_s \left(\begin{matrix} d_1, \dots, d_r \\ \beta_1, \dots, \beta_s \end{matrix} ; z \right) := \sum_{n \geq 0} \frac{(d_1)_n \dots (d_r)_n}{(\beta_1)_n \dots (\beta_s)_n n!} z^n,$$

where $(\alpha)_n := \frac{\Gamma(\alpha+n)}{\Gamma(\alpha)} = \alpha(\alpha+1) \dots (\alpha+n-1)$.

$a = (a_1, \dots, a_r)$, $b = (b_1, \dots, b_s)$, $a_i, b_j \in \mathbb{N}$.

$$u(a, b, z) := \sum_{n \geq 0} u_n(a, b) z^n.$$

Lemma

For a, b , with $r < s$, $\sum_{i=1}^r a_i = \sum_{j=1}^s b_j$,

$\exists d_0, \dots, d_d, \beta_1, \dots, \beta_d \in \mathbb{Q}$ (s.t.) $u(a, b, z) = d_+ F_d \left(\begin{matrix} d_0, \dots, d_d \\ \beta_1, \dots, \beta_d \end{matrix}; Mz \right)$,

where $M := \prod_{i=1}^r a_i / \prod_{j=1}^s b_j$

(proof)

d_i, β_j are obtained by

$$\frac{\prod_{i=1}^d (X - \exp(2\pi i d_i))}{\prod_{j=1}^d (X^{a_j} - 1)} = \frac{\prod_{i=1}^d (X - \exp(2\pi i \beta_i))}{\prod_{j=1}^d (X^{b_j} - 1)}$$

直接計算

$$\beta_0 = 1, d = \left(\sum_{i=1}^r a_i \right) - 1.$$

Theorem [Bober]

Let a, b, L as in page ②.

TFAE.

(i) For $(m_1, \dots, m_{r+s}) = \left(\frac{a_1}{L}, \dots, \frac{a_r}{L}, \frac{b_1}{L}, \dots, \frac{b_s}{L} \right)$, $r \leq s$,

$$f(x) := \sum_{i=1}^r \left\lfloor \frac{x}{m_i} \right\rfloor - \sum_{j=r+1}^{r+s} \left\lfloor \frac{x}{m_j} \right\rfloor \in \{0, 1, \dots, s-r\}$$

(ii) $\sum_{i=1}^r a_i = \sum_{j=1}^s b_j$ & $u(a, b) \in \mathbb{Z}$ (for $n \in \mathbb{N}$)

Moreover, if $s = r+1$, they are also equiv to.

(iii) $u(a, b, z)$ is algebraic / $\mathbb{Q}(z)$

(i.e. u is a root of $F(x) \in \mathbb{Q}(z)[X]$).

(proof)

(ii) $\Leftrightarrow$ (iii) [Rodriguez-Villagas] (2005, preprint)

He reduced (ii) to the result of Beukers-Heckmann's result. (1989)

(i.e. $d_+ F_d(z)$ is algebraic / $\mathbb{Q}(z)$)

(i) $\Rightarrow$ (ii) Omit.

§2

p : prime

$\mathbb{F}_p$: finite field

$w \in \text{Hom}(\mathbb{F}_p^x, \mathbb{C}^x) \simeq (\mathbb{Z}/p\mathbb{Z})^x$: a generator (fix).

$g(w) := \sum_{x \in \mathbb{F}_p} u(x) \exp\left(\frac{2\pi i x}{p}\right) \in \mathbb{C}$: Gauss Sum.

For $m \in \mathbb{Z}$, $g(m) := g(w^m)$

Definition

$d_0, \dots, d_d, \beta_1, \dots, \beta_d \in \mathbb{Q}$. (s.t.) $d_i(p-1), \beta_j(p-1) \in \mathbb{Z}$.

For $x \in \mathbb{F}_p$

$$d_d \mathbb{F}_d \left(\begin{matrix} d_0, \dots, d_d \\ \beta_1, \dots, \beta_d \end{matrix} : x \right)_{\mathbb{F}_p} := \frac{1}{p-1} \sum_{d=0}^{p-2} \prod_{i=0}^d \frac{g(m+d_i(p-1))}{g(d_i(p-1))} \cdot \frac{g(-m-\beta_i(p-1))}{g(-\beta_i(p-1))} w(x)_m$$

$$u(a_1, b : x)_{\mathbb{F}_p} := \sum_{m=0}^{p-2} \frac{(-1)^{p-2}}{1-p} (-1)^{-s(0)+s(m)} \cdot g(a_1 m - b m) w(x)_m$$

Where $g(a_1 m - b m) := g(a_1 m) \dots g(a_r m) g(-b_1 m) \dots g(-b_s m)$
 $S(m) := \text{mult} \exp\left(\frac{2\pi i m}{p-1}\right) gcd\left(\prod_i X^{a_i-1}, \prod_j X^{b_j-1}\right)$

Theorem [K]

a_1, b : given, $s=r+1$

Suppose that $\sum_i a_i = \sum_j b_j$.

Then $u(a_1, b : z)$ is algebraic / $\mathbb{Q}(z)$.

$$\Leftrightarrow \forall p \equiv 1 \pmod L, \quad u(a_1, b : x)_{\mathbb{F}_p} \in \mathbb{Z} \quad \forall x \in \mathbb{F}_p.$$

In particular, for $a_0, a_1, \dots, a_d, \beta_1, \dots, \beta_d \in \mathbb{Q}$

$$(s.t.) \quad \prod_{i=0}^d \frac{X - \exp(2\pi i a_i)}{X - \exp(2\pi i \beta_i)} = \frac{\prod_i X^{a_i} - 1}{\prod_i X^{b_i} - 1}$$

Then

$d+1 F_d \left(\begin{smallmatrix} a_0, \dots, a_d \\ \beta_1, \dots, \beta_d \end{smallmatrix} : z \right)$ is algebraic / $\mathbb{Q}(z)$

$$\Leftrightarrow d+1 F_d \left(\begin{smallmatrix} a_0, \dots, a_d \\ \beta_1, \dots, \beta_d \end{smallmatrix} : z \right)_{\mathbb{F}_p} \in \mathbb{Z} \text{ for } \forall x \in \mathbb{F}_p$$